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COMPREHENSIVE SYSTEM ANALYSIS REPORT

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Project: AG&P Yard
Project Number: AGP-25101-CAL-ELE-E01-0001.00
Engineer: Engr. Benjie Faraon
Date: 11/2/2025, 7:15:13 PM UTC
Software: PwrSys Pro - Short Circuit Analyzer v1.0
Author: Engr. B. P. Faraon
Report Type: Enhanced System Report (Issue #6)

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EXECUTIVE SUMMARY

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PROJECT OVERVIEW:

System Voltage Levels: 13200V, 440V
Total Buses Analyzed: 15
Total Components: 14
Analysis Method: per-unit
Temperature: 75°C
Power Factor: 0.85

SYSTEM HEALTH STATUS:

Overall Status:  ATTENTION REQUIRED

System Metrics:

- Average Fault Current: 23.45 kA
- Maximum Fault Current: 33.56 kA (LCB1-4)
- Average Voltage Drop: 0.38%
- Maximum Voltage Drop: 1.24% (SH1-D16)

Issues Detected:

- Critical Issues: 0 ✓
- High Priority: 3 
- Medium Priority: 6 
- Low Priority: 0 ✓

KEY FINDINGS:

1. ☒ Voltage drop well within recommended limits (max 1.24%)
2. ☒ Fault currents within typical breaker rating capabilities
3. ☒ 15/15 buses compliant with IEEE 141 voltage drop limits

IMMEDIATE ACTIONS REQUIRED:

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SYSTEM LOAD ANALYSIS

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CONNECTED LOAD SUMMARY:

Total Connected Load: 3705.91 A
Total Connected Power: 46506.54 kVA (39530.56 kW @ PF=0.85)
System Power Factor: 0.85
Average Voltage Level: 7245 V

LOAD BREAKDOWN BY VOLTAGE LEVEL:

Voltage(V)	Buses	Total Current(A)	Total Power(kVA)	Utilization
13200	8	1017.62	23265.89	34.3%
440	7	2688.29	2048.75	90.7%
Total		3705.91	46506.54	

LOAD DISTRIBUTION BY BUS TYPE:

Bus Type	Count	Average Load(A)	Total Power(kVA)	Diversity Factor
source 1.0	1	341.17	7800.19	
distribution 1.2	9	231.39	16537.28	
branch 1.3	5	256.44	977.17	

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EQUIPMENT SUMMARY

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TRANSFORMERS (2):

Equipment Tag	Rating(kVA)	Primary(V)	Secondary(V)	Loading(%)
Tap Setting	Status			
XFRMR-LCB1-1000	1000	13200	440	24.4%
0 ✓ OK				
XFRMR-LCA1-1000	1000	13200	440	82.8%
2.5 ⚠ WARN				

Total Capacity: 2000 kVA
Total Load: 1072 kVA (53.6% avg loading)
Spare Capacity: 928 kVA (46.4%)
Status: ✓ Adequate capacity

CABLES (11):

Voltage Level	Count	Total Length(ft)	Avg Size	Material	Parallel Runs
13200	7	2025.3	250	copper	None
440	4	1590.0	500	copper	4x

Total: 11 cables, 3615 ft total length
Estimated Cable Investment: \$54230 (estimated @ \$15/ft avg)
Status: ✓ All cables within thermal limits

MOTORS (1):

Tag/Name	HP	Voltage(V)	FLC(A)	Type	Status
Comp-001	250	440	319.9	induction	✓ OK

Total Motor Load: 250 HP (319.9 A, 243.8 kVA)
Motor Contribution to Fault: ~1919.4 A (6x FLC typical)
Status: ✓ Operating within design parameters

VOLTAGE DROP ANALYSIS - SYSTEM SUMMARY

OVERALL SYSTEM PERFORMANCE:

Worst Case Voltage Drop: 1.24% (SH1-D16)
System Average Drop: 0.38%
IEEE 141 Compliance: ☒ FULLY COMPLIANT
NEC Compliance: ☒ COMPLIANT

Compliance Summary:

- Compliant Buses: 15 of 15 (100.0%)
- Warnings: 0
- Violations: 0

VOLTAGE DROP BY SYSTEM SECTION:

Section	Voltage Drop(%)	Status	Compliance Limit

Primary Feeders (>1kV)	0.07	✓ OK	3% (NEC 215.2)
Transformers	1.66	✓ OK	4% (IEEE 141)
Secondary Feeders (<1kV)	0.74	✓ OK	3% (NEC 215.2)
Combined System	0.38	✓ OK	7% (IEEE 141)

VOLTAGE REGULATION IMPROVEMENT OPPORTUNITIES:

1. Transformer Tap Adjustment:

- Current taps applied: 1 of 2 transformers
- Available range: -5% to +5% (±2.5% steps typical)
- Estimated improvement: 2-5% voltage boost available

2. Cable Upsizing:

- 0 cables could benefit from larger conductors
- Estimated improvement: 0.5% - 1.5% voltage drop reduction per cable
- Cost: Medium to High (cable replacement during maintenance)

3. Load Balancing:

- Redistribute loads among parallel feeders
- Estimated improvement: 0.2% - 0.5% voltage drop reduction
- Cost: Minimal (operational change)

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SHORT CIRCUIT ANALYSIS - SYSTEM SUMMARY

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FAULT CURRENT SUMMARY:

Maximum Fault Current: 33.56 kA @ LCB1-4
Minimum Fault Current: 14.23 kA @ SH1-D16
Average Fault Current: 23.45 kA
Average X/R Ratio: 3.07

System Analysis Method: per-unit

FAULT CURRENT BY VOLTAGE LEVEL:

Voltage Level	Buses	Avg Fault(kA)	Max Fault(kA)	Min Fault(kA)	Avg X/R
MV ($\geq 1\text{kV}$)	8	22.71	26.09	19.03	3.72
LV ($< 1\text{kV}$)	7	24.30	33.56	14.23	2.32

EQUIPMENT INTERRUPTING RATING REQUIREMENTS:

Medium Voltage Circuit Breakers ($\geq 1\text{kV}$):

- Minimum Rating Required: 42 kA sym (25% safety margin)
- Recommended Rating: 45 kA sym (next standard rating)
- Typical Equipment: Medium Voltage VCB/OCB
- Status: ✓ Standard equipment adequate

Low Voltage Circuit Breakers ($< 1\text{kV}$):

- Minimum Rating Required: 42 kA sym
- Recommended Rating: 42 kA sym
- Typical Equipment: MCCB/ACB
- Status: ✓ Standard MCCB adequate

MOTOR CONTRIBUTION:

Total Motor HP: 250 HP
Motor Full Load Current: 319.9 A
Estimated Fault Contribution: 1919.4 A (6× FLC typical)
Decay Time: 0.5 - 1.0 seconds (typical induction motor)
Impact on Protection: ✓ Accounted for in system analysis

CRITICAL PATH ANALYSIS

LONGEST ELECTRICAL PATHS:

Path #1: MERALCO-CCP → SH1-D16

- Total Length: 2021.3 ft
- Voltage Levels: 13200V → 440V
- Components: 7 (6 cables + 1 transformer)
- Total Impedance: 0.017850 Ω (at 440V)
- Voltage Drop: 1.24% (COMPLIANT)
- Fault Current: 14.23 kA
- Status: ✓ Acceptable

- Path #2: MERALCO-CCP → LCB1-1
- Total Length: 1965.3 ft
 - Voltage Levels: 13200V
 - Components: 5 (5 cables)
 - Total Impedance: 0.400539 Ω (at 13200V)
 - Voltage Drop: 0.10% (COMPLIANT)
 - Fault Current: 19.03 kA
 - Status: ✓ Acceptable

- Path #3: MERALCO-CCP → LCB1-4
- Total Length: 1965.3 ft
 - Voltage Levels: 13200V → 440V
 - Components: 6 (5 cables + 1 transformer)
 - Total Impedance: 0.007569 Ω (at 440V)
 - Voltage Drop: 0.86% (COMPLIANT)
 - Fault Current: 33.56 kA
 - Status: ✓ Acceptable

- Path #4: MERALCO-CCP → Motor Bus
- Total Length: 1965.3 ft
 - Voltage Levels: 13200V → 440V
 - Components: 7 (5 cables + 1 transformer)
 - Total Impedance: 0.008189 Ω (at 440V)
 - Voltage Drop: 0.86% (COMPLIANT)
 - Fault Current: 31.02 kA
 - Status: ✓ Acceptable

- Path #5: MERALCO-CCP → SH1-C16
- Total Length: 1871.3 ft
 - Voltage Levels: 13200V → 440V
 - Components: 7 (6 cables + 1 transformer)
 - Total Impedance: 0.015503 Ω (at 440V)
 - Voltage Drop: 0.98% (COMPLIANT)
 - Fault Current: 16.39 kA
 - Status: ✓ Acceptable

MOST CRITICAL COMPONENTS:

Component Risk Level	Criticality	Reason
XFRMR-LCB1-1000 MEDIUM	HIGH	High fault current + loading
XFRMR-LCA1-1000 MEDIUM	HIGH	High fault current + loading
P-13K-DS01-DS02 LOW	MEDIUM	Long run (709ft)
P-440-LCA14-SH1D16 LOW	MEDIUM	Long run (630ft)
P-13K-DS03-LCB11 LOW	MEDIUM	Long run (604ft)

MERALCO-CCP (Utility Feed) HIGH Single point of failure
MEDIUM

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COST IMPACT ANALYSIS

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IMMEDIATE ACTIONS (0-30 DAYS):

Action Timeline	Priority	Est. Cost
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✓ No immediate actions required.

SHORT-TERM ACTIONS (1-6 MONTHS):

Action Timeline	Priority	Est. Cost
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No short-term actions identified.

LONG-TERM IMPROVEMENTS (6-24 MONTHS):

Action Timeline	Priority	Est. Cost
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1. Cable upsizing (5 cables @ 500+ ft) months	MEDIUM	\$50K-150K	12-18
2. Power factor correction months	MEDIUM	\$15K-30K	6-12
3. Monitoring system installation months	MEDIUM	\$10K-25K	3-6

Subtotal Long-Term Improvements: \$75,000-\$205,000

TOTAL ESTIMATED COST (ALL RECOMMENDATIONS): \$75,000-\$205,000

COST AVOIDANCE THROUGH DIVERSITY FACTORS:

By applying IEEE 141-1993 diversity factors, the system design can realize:

- Load Reduction: 20-30% through diversity (typical for industrial systems)
- Transformer Capacity Saved: ~500 kVA (reduced oversizing)
- Cable Size Reduction: ~2 AWG sizes smaller than without diversity

- Estimated Capital Savings: \$50,000 - \$100,000
- ROI: Immediate (design phase savings - no additional cost)

Note: Diversity factors are already applied in this analysis per Feature #5.

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STANDARDS COMPLIANCE DETAILED ANALYSIS

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NEC 2023 COMPLIANCE:

Article 210.19(A) - Branch Circuit Conductors:

- ✓ Compliant Buses: 5 of 5 branch circuits
- Limit: 5% maximum voltage drop
- Worst Case: 1.24%
- Status: ✓ COMPLIANT

Article 215.2(A)(1) - Feeder Conductors:

- ✓ Compliant Buses: 9 of 10 feeders
- Limit: 3% maximum voltage drop (recommended)
- Worst Case: 0.86%
- Status: ⚠ REVIEW REQUIRED

Article 220 - Branch-Circuit, Feeder, and Service Load Calculations:

- ✓ Demand factors applied per NEC tables
- ✓ Method: IEEE 141 diversity factors (more conservative than NEC minimum)
- Status: ✓ Meets or exceeds NEC requirements

Article 430.24 - Motor Load Calculations:

- ✓ Motor FLC used with 125% multiplier for continuous duty
- ✓ Motor contribution to fault current accounted for
- Status: ✓ Proper motor load calculations applied

IEEE 141-1993 (RED BOOK) COMPLIANCE:

Section 3.4 - Voltage Drop:

- ✓ Compliant Buses: 15 of 15
- System Limit: 7% maximum (combined feeder + branch)
- Method: Component-by-component with voltage tracking
- Worst Case: 1.24%
- Status: ☑ FULLY COMPLIANT

Section 5 - Short Circuit Analysis:

- ✓ Point-to-point method applied
- ✓ Motor contribution included per Section 5.3
- ✓ X/R ratios calculated and verified
- Status: ✓ Proper short circuit methodology

Table 3-5 - Diversity Factors:

- ✓ Applied 1.2-1.3 diversity factors per bus type
- ✓ Result: 20-30% load reduction (typical)
- Status: ✓ Conservative application of diversity

IEEE C57.12.00 - TRANSFORMER STANDARDS:

Section 5.3 - Short-Time Overload Capability:

- ✓ COMPLIANT
- Transformers Analyzed: 2
- Overloaded: 0
- Action: All transformers operating within ratings
- Timeline: N/A

Section 7 - Temperature Rise:

- ✓ All transformers < 100% rated load
- Average Loading: 53.6%
- Status: ✓ Normal operating conditions

IEEE 242-2001 (BUFF BOOK) - PROTECTION:

Chapter 3 - Protective Device Coordination:

- ⚠ REVIEW REQUIRED
- Motor Contribution: Present - may affect coordination
- Action: Update time-current curves to account for motor contribution
- Timeline: 60 days

Chapter 6 - Transformer Protection:

- ✓ Primary and secondary protection recommended
- Status: ✓ Adequate protection scheme assumed

SYSTEM EFFICIENCY METRICS

POWER QUALITY INDICATORS:

System Power Factor: 0.85 (ACCEPTABLE)

- Target: 0.90 or higher
- Improvement Opportunity: 0.05 (5.0%)
- Method: Power factor correction capacitors
- Estimated Cost: \$15,000 - \$30,000
- Annual Savings: \$3,000 - \$6,000 (reduced demand charges)
- Payback Period: 5-10 years

Total Harmonic Distortion (THD): Not measured

- Recommendation: Conduct power quality study
- Typical Acceptable Limit: < 5% (IEEE 519)
- Cost: \$5,000 - \$10,000

Voltage Unbalance: Not measured

- Recommendation: Monitor phase voltages
- Acceptable Limit: < 2% (NEMA MG-1)
- Impact: Motor derating, reduced efficiency

SYSTEM LOSSES:

Cable I²R Losses:

- MV System (≥1kV): ~4.1 kW
- LV System (<1kV): ~6.1 kW
- Total Cable Losses: ~10.2 kW

Transformer Losses:

- No-Load Losses: ~6.0 kW (2 transformers)
- Load Losses: ~11.2 kW @ current loading
- Total Transformer Losses: ~17.2 kW

Total System Losses: ~27.3 kW (0.13% of total load)

Annual Energy Loss: 239543 kWh

Annual Cost @ \$0.12/kWh: \$28745

Improvement Opportunity: Reduce losses by 0.5% = \$113096/year

ENERGY EFFICIENCY RECOMMENDATIONS:

1. Power Factor Correction

- Investment: \$15,000 - \$30,000
- Annual Savings: \$3,000 - \$6,000 (reduced demand charges)
- Payback: 5-10 years
- Priority: MEDIUM

2. Transformer Upgrade to Higher Efficiency Units

- Investment: \$60,000 - \$120,000
- Annual Savings: \$4,000 - \$8,000 (reduced losses)
- Payback: 15-20 years (long-term)
- Priority: LOW (consider during replacement cycle)

3. Cable Upsizing (reduce I²R losses)

- Investment: \$50,000 - \$150,000
- Annual Savings: \$5,000 - \$10,000
- Payback: 10-15 years
- Priority: MEDIUM (combine with voltage drop improvements)

4. Load Balancing

- Investment: \$2,000 - \$5,000
- Annual Savings: \$1,000 - \$2,000
- Payback: 2-5 years
- Priority: HIGH (RECOMMENDED - Quick win)

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MAINTENANCE RECOMMENDATIONS

PREVENTIVE MAINTENANCE SCHEDULE:

Monthly:

- ☐ Visual inspection of transformers for oil leaks, unusual sounds
- ☐ Check for abnormal heating at all connections
- ☐ Verify cooling fan operation (if applicable)
- ☐ Inspect for signs of moisture, corrosion, or wildlife intrusion
- ☐ Review load data from monitoring systems (if installed)

Quarterly:

- ☐ Thermographic inspection of all electrical connections
- ☐ Measure and record phase currents at all distribution points
- ☐ Check transformer oil level and temperature
- ☐ Inspect cable terminations and splices
- ☐ Test ground fault protection systems
- ☐ Clean and inspect switchgear, MCCs, and panelboards

Semi-Annually:

- ☐ Transformer oil analysis (dissolved gas analysis - DGA)
- ☐ Insulation resistance testing (Megger) on cables
- ☐ Ground resistance testing
- ☐ Battery backup system testing
- ☐ Verify voltage drop measurements at critical loads
- ☐ Review and update protective relay settings

Annually:

- ☐ Comprehensive transformer inspection
- ☐ Breaker contact resistance testing
- ☐ Protective relay testing and calibration
- ☐ Arc flash label verification and update
- ☐ Single-line diagram update with any system changes
- ☐ Emergency shutdown and backup power testing
- ☐ Review and update maintenance logs

Every 3-5 Years:

- ☐ Transformer internal inspection and BIL testing
- ☐ Cable hi-pot testing (insulation integrity)
- ☐ Major breaker overhaul or replacement
- ☐ Ground grid integrity testing
- ☐ Power quality study (harmonics, voltage variations)
- ☐ Short circuit study update
- ☐ Load flow study update
- ☐ Arc flash hazard analysis update

MONITORING RECOMMENDATIONS:

1. Install Power Quality Meters at Main Feeders

- Cost: \$5,000 - \$10,000
- Benefits:
 - Real-time monitoring of voltage, current, power factor
 - Early fault detection
 - Load trending and forecasting
 - Energy consumption tracking
- ROI: 3-5 years (reduced downtime, optimized maintenance)
- Priority: HIGH

2. Transformer Temperature Monitoring

- Cost: \$3,000 - \$6,000 per transformer
- Benefits:
 - Overload protection
 - Predictive maintenance (detect cooling system failures)
 - Extend transformer life through optimized loading
- ROI: 2-3 years (prevent catastrophic failures)
- Priority: MEDIUM (install on critical transformers first)

3. Motor Condition Monitoring

- Cost: \$2,000 - \$4,000 per motor
- Benefits:
 - Early bearing failure detection
 - Winding insulation monitoring
 - Vibration analysis
 - Energy efficiency tracking
- ROI: 1-2 years (prevent unplanned downtime)
- Priority: MEDIUM (install on critical process motors)

4. Automated Load Management System

- Cost: \$15,000 - \$30,000
- Benefits:
 - Automatic load shedding during peak demand
 - Demand response participation
 - Energy cost optimization
 - Real-time diversity factor verification
- ROI: 2-4 years (reduced demand charges)
- Priority: LOW (consider for future expansion)

SPARE PARTS RECOMMENDATIONS:

Critical Spare Parts to Maintain:

- ☐ Circuit breakers (common frame sizes)
- ☐ Motor starters and contactors
- ☐ Control power transformers
- ☐ Fuses (all ratings used in system)
- ☐ Cable connectors and lugs (common sizes)
- ☐ Ground fault and arc fault relays
- ☐ Emergency lighting battery packs
- ☐ Transformer oil (for oil-filled units)
- ☐ Cable pulling lubricant and splicing materials

Recommended Stock Levels:

- Molded Case Circuit Breakers: 1-2 per common frame size
- Motor Starters: 1 per common HP rating (critical motors)
- Control Transformers: 2-3 units (various VA ratings)
- Fuses: 10-20 per rating (high-usage ratings)
- Cable Lugs/Connectors: 20-50 per common size

CONCLUSION AND NEXT STEPS

OVERALL SYSTEM ASSESSMENT:

The electrical distribution system is generally acceptable but has areas requiring attention per IEEE and NEC standards.

System Rating: NEEDS ATTENTION

Key Strengths:

- ✓ Excellent voltage regulation (avg 0.38% < 3%)
- ✓ Fault currents within standard breaker ratings
- ✓ 100% IEEE 141 voltage drop compliance
- ✓ Adequate cable sizing for thermal limits

AREAS REQUIRING ATTENTION:

- ⚠ 3 High Priority Issues requiring attention within 30 days
- ⚠ 6 Medium Priority issues - plan for next maintenance cycle
- ⚠ Power factor below optimal (0.85) - consider correction

RECOMMENDED PRIORITY ACTIONS:

Priority 1 (0-30 days) - IMMEDIATE:

- ✓ No immediate critical actions required

Priority 2 (1-6 months) - SHORT-TERM:

Priority 3 (6-24 months) - LONG-TERM:

- Cable upsizing for long runs (>500 ft)
- Power factor correction evaluation
- Transformer efficiency upgrade (during replacement)
- Develop 5-year capital improvement plan

ESTIMATED INVESTMENT:

Immediate Actions (Priority 1): \$0

Short-Term Actions (Priority 2): \$20K-50K

Long-Term Improvements (Priority 3): \$100K-250K

Total Program Cost: \$110K-300K

Note: Costs are estimates and may vary based on:

- Local labor rates and material costs
- System accessibility and outage requirements
- Specific equipment manufacturer selections
- Regulatory compliance requirements

EXPECTED BENEFITS:

Immediate Benefits (0-12 months):

- Improved system reliability and safety
- Reduced risk of equipment failure
- Regulatory compliance maintained
- Enhanced worker safety (arc flash mitigation)

Operational Benefits (1-5 years):

- Lower maintenance costs (predictive vs reactive)
- Reduced unplanned downtime
- Extended equipment life
- Improved power quality

Financial Benefits (5-10 years):

- Energy cost savings: \$10,000 - \$20,000/year (potential)
- Avoided replacement costs from premature failures
- Reduced insurance premiums (improved safety)
- Increased system capacity for future growth

FINAL RECOMMENDATIONS:

Based on this comprehensive analysis, the following approach is recommended:

1. IMMEDIATE (Within 30 days):

- Implement monitoring for continuous system health assessment
- Verify all circuit breaker ratings meet fault current requirements
- Update arc flash labels if system changes have occurred

2. SHORT-TERM (3-6 months):

- Conduct detailed cable sizing review for voltage drop optimization
- Implement load balancing strategies
- Install power quality monitoring at main feeders
- Review and update protective device coordination study

3. LONG-TERM (1-3 years):

- Develop 5-year electrical infrastructure capital plan
- Evaluate power factor correction economics
- Plan for cable upsizing during maintenance outages
- Consider energy efficiency upgrades (LED lighting, VFDs, etc.)

4. ONGOING:

- Maintain preventive maintenance schedule per recommendations
- Monitor load growth and diversity factors
- Update electrical studies every 3-5 years
- Train staff on proper operation and emergency procedures

SUMMARY OF ALL BUSES

Bus Name Demand(A) Status	Voltage(V)	Fault(kA)	X/R Ratio	VDrop(%)

MERALCO-CCP	13200	26.09	5.99	0.00
341.17 ✓ OK				
EHV	13200	25.73	5.70	0.02
401.57 ✓ OK				
DS-01	13200	24.03	4.00	0.06
90.57 ✓ OK				
LCJ1-1	13200	23.82	3.91	0.06
43.70 ✓ OK				
DS-02	13200	21.39	2.77	0.09
46.87 ✓ OK				
DS-03	13200	20.91	2.63	0.10
46.87 ✓ OK				
LCA1-1	13200	20.73	2.60	0.10
36.21 ✓ OK				
LCA1-4	440	31.16	2.89	0.17
1086.20 ⚠ HIGH				
SH1-A16	440	24.42	2.25	0.40
257.70 ⚠ MEDIUM				
SH1-B16	440	19.29	1.91	0.69
263.14 ⚠ MEDIUM				
SH1-C16	440	16.39	1.75	0.98
281.45 ⚠ MEDIUM				
SH1-D16	440	14.23	1.65	1.24
283.91 ⚠ MEDIUM				
LCB1-1	13200	19.03	2.18	0.10
10.66 ✓ OK				
LCB1-4	440	33.56	2.94	0.86
319.89 ⚠ HIGH				
Motor Bus	440	31.02	2.86	0.86
196.00 ⚠ HIGH				

CABLE TAG DIRECTORY:

Tag	Description Size/Material	From	To
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P-13K-MECO-EHV1	Service Feeder from CCP	MERALCO	EHV
	500 copper		
P-13K-EHV4-DS01	MV Feeder 2 - DS01	EHV	
DS-01	250 copper		
P-13K-DS01-DS02	MV Feeder 2 - DS02	DS-01	
DS-02	250 copper		
P-13K-DS01-LCJ11	ML Branch Feeder - SS LCJ1	DS-01	
LCJ1-1	4/0 copper		
P-13K-DS02-DS03	MV Feeder 2 - DS03	DS-02	
DS-03	250 copper		
P-13K-DS03-LCA11	ML Branch Feeder - SS LCA1	DS-03	
LCA1-1	250 copper		
P-440-LCA14-SH1-A16	LV Branch Feeder - DB SH1-A16	LCA1-4	
SH1-A16	500 copper (3x)		
P-440-LCA14-SH1-B16	LV Branch Feeder - DB SH1-B16	LCA1-4	
SH1-B16	500 copper (3x)		
P-440-LCA14-SH1C16	LV Branch Feeder - DB SH1-C16	LCA1-4	
SH1-C16	500 copper (3x)		
P-440-LCA14-SH1D16	LV Branch Feeder - DB SH1-D16	LCA1-4	
SH1-D16	500 copper (3x)		
P-13K-DS03-LCB11	MV Branch Feeder - SS LCB1	DS-03	
LCB1-1	250 copper		

SYSTEM-WIDE RECOMMENDATIONS:

Total Recommendations: 9

- Critical Issues: 0
- High Priority: 3
- Medium Priority: 6
- Low Priority: 0

RECOMMENDATIONS BY CATEGORY:

Transformer: 3

Cable: 5

Protection Coordination: 1

ALL RECOMMENDATIONS BY BUS:

BUS: LCA1-4 (440V)

1. [HIGH] Transformer Secondary Fault High

Category: Transformer

Finding: High secondary fault current - verify transformer thermal capability

Action: Check transformer: 1) Thermal withstand time, 2) Primary protection settings, 3) Secondary protection coordination.

Impact: Potential transformer thermal damage

Cost: MEDIUM | Effort: Low (verification and protection adjustment)

Standard: IEEE C57.12.00, IEEE 242

2. [MEDIUM] Long Cable Run High Drop

Category: Cable

Finding: Long cable run (>500 ft) with voltage drop >2%

Action: Consider: 1) Larger conductor size, 2) Parallel cables, 3)

Intermediate voltage boost.

Impact: Poor voltage regulation

Cost: MEDIUM | Effort: High (cable replacement)

Standard: IEEE 141-1993

BUS: SH1-A16 (440V)

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Action: Consider: 1) Larger conductor size, 2) Parallel cables, 3)

Intermediate voltage boost.

Impact: Poor voltage regulation

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Standard: IEEE 141-1993

BUS: SH1-B16 (440V)

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Standard: IEEE 141-1993

BUS: SH1-C16 (440V)

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Action: Consider: 1) Larger conductor size, 2) Parallel cables, 3)

Intermediate voltage boost.

Impact: Poor voltage regulation
Cost: MEDIUM | Effort: High (cable replacement)
Standard: IEEE 141-1993

BUS: SH1-D16 (440V)

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Category: Cable

Finding: Long cable run (>500 ft) with voltage drop >2%

Action: Consider: 1) Larger conductor size, 2) Parallel cables, 3)

Intermediate voltage boost.

Impact: Poor voltage regulation

Cost: MEDIUM | Effort: High (cable replacement)

Standard: IEEE 141-1993

BUS: LCB1-4 (440V)

1. [HIGH] Transformer Secondary Fault High

Category: Transformer

Finding: High secondary fault current - verify transformer thermal capability

Action: Check transformer: 1) Thermal withstand time, 2) Primary protection settings, 3) Secondary protection coordination.

Impact: Potential transformer thermal damage

Cost: MEDIUM | Effort: Low (verification and protection adjustment)

Standard: IEEE C57.12.00, IEEE 242

BUS: Motor Bus (440V)

1. [HIGH] Transformer Secondary Fault High

Category: Transformer

Finding: High secondary fault current - verify transformer thermal capability

Action: Check transformer: 1) Thermal withstand time, 2) Primary protection settings, 3) Secondary protection coordination.

Impact: Potential transformer thermal damage

Cost: MEDIUM | Effort: Low (verification and protection adjustment)

Standard: IEEE C57.12.00, IEEE 242

2. [MEDIUM] Motor Contribution Significant

Category: Protection Coordination

Finding: Motor contribution to fault current detected

Action: Ensure protective device coordination accounts for motor contribution. Review time-current curves.

Impact: Protection may not coordinate properly

Cost: LOW | Effort: Low (protection study update)

Standard: IEEE 242-2001 Buff Book

IEEE STANDARDS COMPLIANCE SUMMARY

⚠ COMPLIANCE ISSUES DETECTED

The following buses require attention:

- LCA1-4: 1 High Priority
- LCB1-4: 1 High Priority
- Motor Bus: 1 High Priority

REPORT PREPARED BY

Engineer: Engr. Benjie Faraon
Software: PwrSys Pro - Short Circuit Analyzer v1.0
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Standards Applied:

- ✓ NEC 2023 (NFPA 70)
- ✓ IEEE 141-1993 (Red Book - Industrial Power Systems)
- ✓ IEEE 242-2001 (Buff Book - Protection & Coordination)
- ✓ IEEE C57.12.00 (Transformer Standards)
- ✓ IEEE 519 (Harmonic Control)
- ✓ API RP 540 (Petroleum Facilities)
- ✓ NEMA MG-1 (Motors and Generators)

This report represents a comprehensive analysis of the electrical distribution system based on provided data and industry-standard calculation methods. Field verification and manufacturer specifications should be confirmed for critical components before implementation.

END OF COMPREHENSIVE SYSTEM REPORT